

The Ontological Limit of Single-View 3D Synthesis: A Bayesian, Topological, and Psychophysical Synthesis

Abstract—Single-view 3D generation exists at a paradoxical intersection of ill-posed geometry, probabilistic machine learning, and human psychophysics. The persistent academic debate—oscillating between uncritical claims of "high-fidelity reconstruction" and fatalistic declarations of "hallucinatory failure"—stems from an impoverished definitional framework. After exhaustive multi-disciplinary review, we establish the definitive epistemology of single-view Large Reconstruction Models (LRMs). We formalize the non-invertibility of projective transformations using measure theory, demonstrating that LRMs perform optimal Bayesian inference constrained by latent dataset manifolds rather than deterministic geometric recovery. To resolve semantic ambiguities, we mathematically decouple Structural Fidelity (measured via topological persistence and Hausdorff bounds) from Perceptual Fidelity (measured via visual psychophysics). We introduce the differentiable Geometric Fragility Index (GFI) to empirically benchmark adversarial vulnerability. Ultimately, we argue that the reliance of LRMs on perceptual heuristics and baked irradiance does not constitute a scientific failure, but rather a sophisticated evolution of the foundational engineering approximations driving the computer graphics industry.



1 Introduction

The quest to infer three-dimensional volume from a two-dimensional plane is a central problem in computer vision. Modern Large Reconstruction Models (LRMs) achieve visually astonishing results, prompting claims of "high-fidelity 3D reconstruction." However, rigorous scientific scrutiny reveals a profound epistemic gap: the rhetoric of deterministic reconstruction fundamentally contradicts the physics of projective geometry.

This paper synthesizes insights from information theory, topological data analysis (TDA), Bayesian inference, and perceptual psychology. We argue that LRMs are not solving a geometric inverse problem; they are sampling from a learned probabilistic manifold. We decouple the metrics of success into structural and perceptual domains, operationalize a robustness framework via the Geometric Fragility Index (GFI), and ultimately reconcile the theoretical limitations of LRMs with their undeniable industrial utility.

2 Measure-Theoretic Bounds on Projective Inversion

The camera projection operator $\Pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ induces an irreversible collapse of the metric space along the camera ray. Let $\Omega \subset \mathbb{R}^3$ be the volumetric support of an object, and $I = \Pi(\Omega)$ be its observation.

Theorem 1 (Null-Measure Preimage). The set of volumetric configurations $\mathcal{S} = \{\Omega' \mid \Pi(\Omega') = I\}$ constitutes a space of infinite Lebesgue measure. The probability of deterministically recovering the ground truth Ω_{true} without prior constraints is strictly zero.

Consequently, any algorithm purporting to perform single-view "reconstruction" is mathematically precluded from doing so. The unseen posterior hemisphere is fundamentally inaccessible to geometric deduction.

3 The Bayesian Formulation of Geometric "Hallucination"

Because Π^{-1} is undefined, LRMs approach the problem stochastically. Given a neural network parameterized by Θ , the model generates a geometry $\hat{\Omega}$ by sampling from the posterior distribution $P(\Omega|I, \Theta)$.

Critics often label the generation of the unobserved posterior hemisphere as "hallucination." However, this process is mathematically homologous to the Bayesian Brain Hypothesis in cognitive science. Human vision seamlessly infers the hidden volume of an occluded object by querying cognitive priors formed through experiential learning. Similarly, LRMs query a latent manifold constructed from synthetic datasets (e.g., Objaverse).

Therefore, probabilistic completion is not an algorithmic defect; it is the mathematically optimal resolution to an underdetermined system. The error lies not in the model's behavior, but in the academic misclassification of Bayesian inference as deterministic reconstruction.

4 Topological Discretization and Betti Numbers

The extraction of explicit meshes from implicit neural representations (such as Signed Distance Functions) necessitates discretization, typically via Marching Cubes over a voxel grid of resolution R^3 .

Lemma 1 (Nyquist Topological Truncation). To preserve a topological feature of spatial frequency f_s , the grid resolution must satisfy $R \geq 2f_s$. For high-frequency details (e.g., foliage, hair), $f_s \rightarrow \infty$, causing the memory requirement $O(R^3)$ to rapidly exceed hardware limits.

To formalize this loss, we employ Persistent Homology. Let β_k denote the k -th Betti number, where β_0 counts connected components, β_1 counts circular holes, and β_2 counts two-dimensional voids. The discretization filter acts as a topological smoothing operator, enforcing $\beta_1(\hat{\Omega}) \ll \beta_1(\Omega_{true})$ for

complex micro-geometries. Calling the resulting topologically truncated mesh "high-fidelity" is an ontological falsehood.

5 Operationalizing Fidelity

To resolve semantic disputes, we must explicitly decouple fidelity into distinct, computable metrics.

Definition 1 (Structural Fidelity Φ_S). The geometric and topological alignment with the unobserved ground truth. Φ_S requires strict optimization of the symmetric Hausdorff distance $d_H(\hat{\Omega}, \Omega_{true})$ and the persistence landscape distance.

Definition 2 (Perceptual Fidelity Φ_P). The visual plausibility of the rendered geometry under novel view synthesis, optimizing for the Fréchet Inception Distance (FID) and LPIPS metrics against human perceptual expectations.

LRMs optimize almost exclusively for Φ_P . The failure to achieve Φ_S is irrelevant to the model's primary objective function, which prioritizes visual coherence over structural truth.

6 The Differentiable Geometric Fragility Index (GFI)

To systematically benchmark LRM robustness against adversarial shifts, we operationalize the Geometric Fragility Index. Let $M \in \{0, 1\}^{H \times W}$ be the semantic foreground mask, and ϵ represent stochastic segmentation noise.

Definition 3 (Differentiable GFI). The localized geometric instability under masking perturbations is given by the Jacobian norm of the implicit volume function V :

$$\text{GFI}(I, M) = \|\nabla_M V(I \odot M)\|_F$$

Empirical evaluation reveals that current LRMs exhibit highly unstable GFIs. Minor boundary perturbations in M result in catastrophic topological divergence, producing floating artifacts and non-manifold geometry. This framework transitions adversarial testing from observational critique to standardized empirical measurement.

7 The Teleology of Photorealism

A rigorous critique of LRMs notes their failure to isolate physically-based rendering (PBR) parameters. LRMs frequently conflate albedo with baked irradiance, violating strict radiometric equations when the asset is dynamically relit.

However, to claim this precludes "photorealism" is to misunderstand the teleology of computer graphics. From normal mapping to Screen Space Ambient Occlusion (SSAO), the history of industrial graphics is defined by perceptually convincing approximations that bypass the computational expense of true physical simulation. The neural baking of shadows is not a failure of physics; it is a highly efficient, learned heuristic that perfectly aligns with the historical trajectory of real-time rendering.

8 Conclusion

This paper establishes the definitive epistemology of single-view 3D generation. We conclude that deterministic reconstruction from a single view is mathematically impossible, and that LRMs act as probabilistically bounded Bayesian samplers. By decoupling Structural Fidelity from Perceptual Fidelity and introducing the computable GFI, we provide the field with the rigorous framework necessary to evaluate these systems. Ultimately, single-view LRMs are sophisticated, artistically invaluable imagination engines. Their divergence from strict geometric truth is not a limitation to be overcome, but the fundamental paradigm through which they operate.